# Mauricio Angel Treviño IV

Houston, TX | Portfolio | linkedin.com/in/mauricio-trevino4/ | maurit2001@gmail.com | U.S. Citizen

## EDUCATION

**Rice University**, Houston, TX — August 2025 – Present
*Master of Science in Space Studies, Professional Science Masters Program, Graduating December 2026* — GPA: 4.0/4.0

**Texas A&M University**, College Station, TX — August 2020 – May 2025
*Bachelor of Science in Mechanical Engineering, Engineering Honors, Summa Cum Laude* — GPA: 3.9/4.0

## PROFESSIONAL EXPERIENCE

**The Exploration Company,** Houston, TX — January 2026 – August 2026
*Crew Systems Engineering Intern - Mechanical/Structures*

- Hand built lab infrastructure for the new Rapid Innovation Lab facility from the ground up
- Developed the full interior of a full-scale Nyx Earth crew vehicle mockup alongside astronaut Dr. Lee Morin
- Modified COTS racing seats with a custom base for crew seating, designed a triple-touchscreen crew control panel, and custom spacecraft control devices; managed assemblies with 500+ components
- Presented the mockup at the RIL grand opening; recipient of the Spirit of Exploration Award at summer all-hands

**Robotics and Automation Design (RAD) Lab,** College Station, TX — May 2024 – July 2025
*Undergraduate Research Assistant*

- Engineered a deployable sensor package launched from RoboBall III to extend mission range, provide GPS localization, and enable visual reconnaissance
- Developed and integrated key subsystems for RoboBall II, including a wireless inductive charging module, real-time data visualization via ROS2/MATLAB, and explored soft-shelled mobility platforms for diverse environments; Co-authored 2 published works, viewable on my ORCiD page: 0009-0002-2986-3486

**NASA Pathways Co-Op,** Stennis Space Center, MS — January 2023 - August 2023
*Rocket Propulsion Test Engineer Co-Op* — January 2024 - May 2024

- Designed and analyzed cryogenic piping and instrumentation systems for rocket propulsion testing, adhering to ASME standards and NASA protocols
- Conducted liquid oxygen transfers and supported test stand integration using model-based engineering approaches
- Evaluated novel valve performance, determining flow coefficients and assessing system-level impacts
- Selected as NASA PAXC Stennis Representative; presented technical work at the National Intern Symposium

## DESIGN TEAM EXPERIENCE

**Society of Automotive Engineers Formula Internal Combustion,** College Station, TX — May 2024 – May 2025
*Aerodynamics Sub-Team Member*

- Tuned event-specific setups using simulation and track validation, contributing to 2nd place overall at the 2025 FSAE Michigan competition
- Led aerodynamic development through full-car CFD simulations in Ansys Fluent, optimizing interactions between wings, undertray, and bodywork for performance gains
- Directed cross-subteam collaboration between aerodynamics, suspension, powertrain, and chassis to ensure cohesive vehicle integration and maximize overall performance
- Manufactured the full aerodynamics package using carbon fiber epoxy composites

**Society of Automotive Engineers Aerospace Design,** College Station, TX — April 2021 – May 2024
*Product Owner of Structures and Material Science Sub-Team*

- Optimized the internal truss structure of an RC aircraft using FEA and hand calculations
- Constructed the aircraft using precision fabrication techniques including custom jig assemblies
- Led the Micro Class team to 1st place overall at the 2024 SAE Aero Design West Competition

## SKILLS & INTERESTS

**Software:** Solidworks (CSWA), Siemens NX, Creo, Fusion 360, ANSYS, MATLAB, LabView, Multisim, Python
**Technical:** DFMA, FEA, CNC Mill & Lathe (ACE, MasterCAM), 3D Printing, Machining, Composite Layup, Soldering, Arduino, Model Based Engineering, Rapid Prototyping, Agile Methodologies, Project Management
**Interests:** Extraterrestrial Exploration, Lander Systems, Propulsion, Crew Environments, Spikeball, FPV Drones